



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 wide notification number F-9-12/2001-U-3 of Government of India)

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Computer Networks/0705210503

Prof. Soubhagya Ranjan Mallick
Assistant Professor (Selection Grade), CSE Dept.
SIT, Nagpur



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The Link layer:

Link layer basics, error detection and correction, parity, checksum and CRC, random access control protocols, taking turns protocols, DOCSIS, Link virtualization, Multiprotocol Label Switching



INTRODUCTION

- In the OSI model, the data link layer is the 2nd layer from the bottom.
- It is responsible for transmitting frames from one node to the next node.
- The main responsibility of the Data Link Layer is to transfer the datagram across an individual link.
- An important characteristic of a Data Link Layer is that a datagram can be handled by different link layer protocols on different links in a path.



INTRODUCTION

- The other responsibilities of this layer are
 - **Framing** - Divides the stream of bits received into data units called frames.
 - **Physical addressing** – If frames are to be distributed to different systems on the same network, data link layer adds a header to the frame to define the sender and receiver.
 - **Flow control**- If the rate at which the data are absorbed by the receiver is less than the rate produced in the sender ,the Data link layer imposes a flow control mechanism.
 - **Error control**- Used for detecting and retransmitting damaged or lost frames and to prevent duplication of frames. This is achieved through a trailer added at the end of the frame.
 - **Medium Access control** - Used to determine which device has control over the link at any given time.



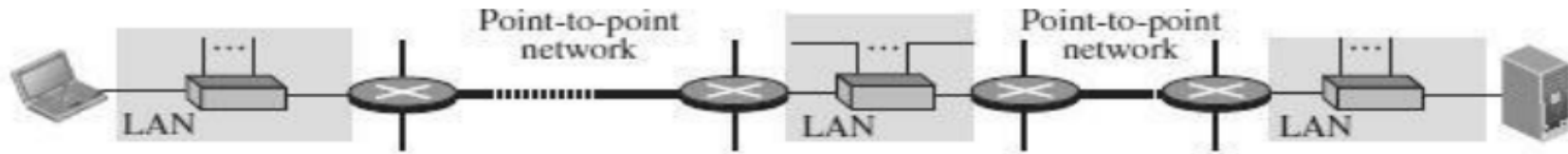
INTRODUCTION

Nodes and Links

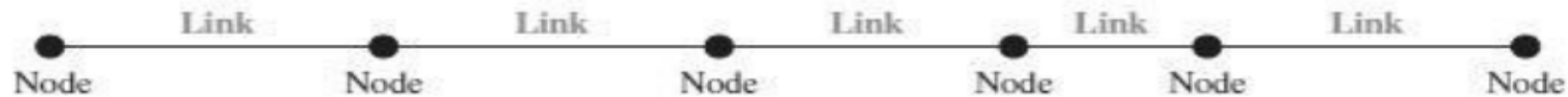
- Communication at the data-link layer is node-to-node.
- The communication channel that connects the adjacent nodes is known as links, and in order to move the datagram from source to the destination, the datagram must be moved across an individual link.
- A data unit from one point in the Internet needs to pass through many networks (LAN and WAN) to reach another point.
- These LANs and WANs are connected by routers.
- The two end hosts and the routers are *nodes* and the networks in-between are *links*.



INTRODUCTION



a. A small part of the Internet



b. Nodes and links

- The first node is the source host; the last node is the destination host.
- The other four nodes are four routers.
- The first, the third, and the fifth links represent the three LANs; the second and the fourth links represent the two WANs.

Two Categories of Links

Point- to-Point link and Broadcast link.

- In a point-to-point link, the link is dedicated to the two devices
- In a broadcast link, the link is shared between several pairs of devices.



2. LINK-LAYER ADDRESSING

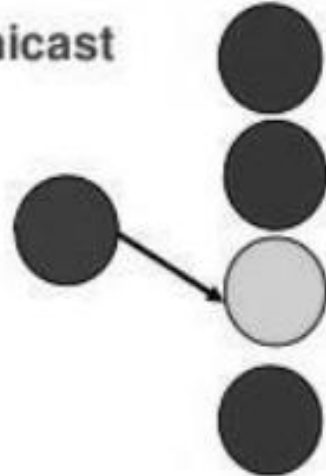
- A link-layer address is sometimes called a link address, sometimes a physical address, and sometimes a MAC address.
- Since a link is controlled at the data-link layer, the addresses need to belong to the data-link layer.
- When a datagram passes from the network layer to the data-link layer, the datagram will be encapsulated in a frame and two data-link addresses are added to the frame header.
- These two addresses are changed every time the frame moves from one link to another.



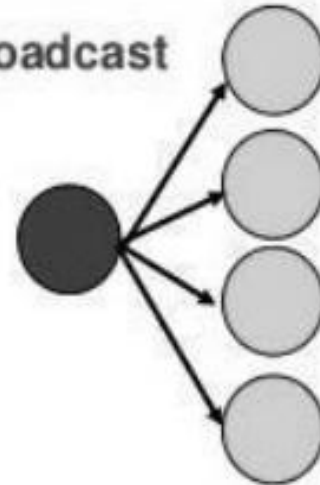
THREE TYPES OF ADDRESSES

The link-layer protocols define three types of addresses: unicast, multicast, and broadcast.

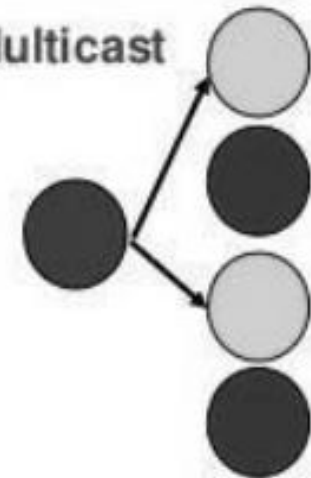
Unicast



Broadcast



Multicast





Unicast Address :

Each host or each interface of a router is assigned a unicast address. Unicasting means one-to-one communication. A frame with a unicast address destination is destined only for one entity in the link.

Multicast Address :

Link-layer protocols define multicast addresses. Multicasting means one-to-many Communication but not all.

Broadcast Address :

Link-layer protocols define a broadcast address. Broadcasting means one-to-all communication. A frame with a destination broadcast address is sent to all entities in the link.